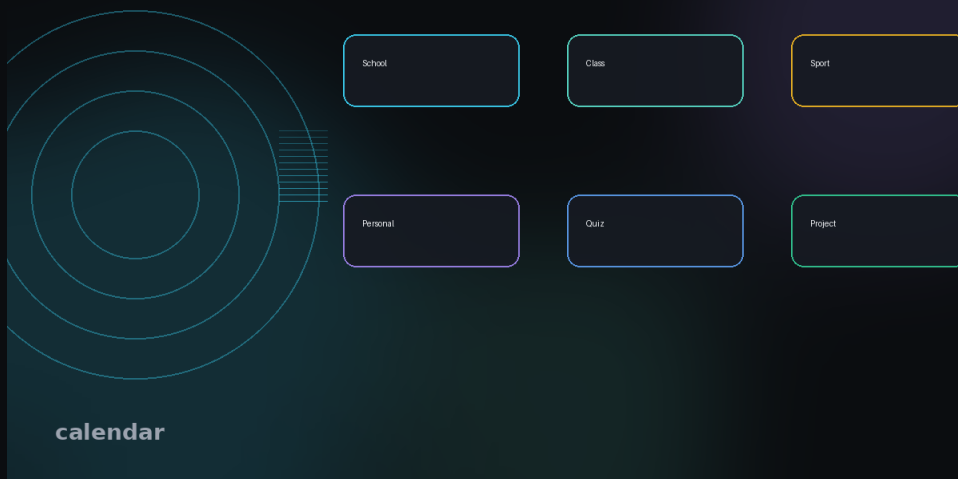




Calendar — Research + Recommended Features

Dark visual edition • 2026-09-24

Card t_908912d4 • CAL • Research digest (no app implementation)

Chuck Broaddus / Kelyra • Confidential

Source: dated research digests under notes/company • graphics illustrate locked recommendations.

Executive recommendation

Locked recommendation

Ship category filters for the full CEO dated taxonomy, end-to-end draft≠publish (pop-quiz law), hat/twin visibility walls, and AI NL search/add over the already-visible set — on the existing Calendar surface (not a greenfield app).

Decision cards

P0 focus Filters · Publish · Hats	Stance Not greenfield
AI NL search + add	Card t_908912d4

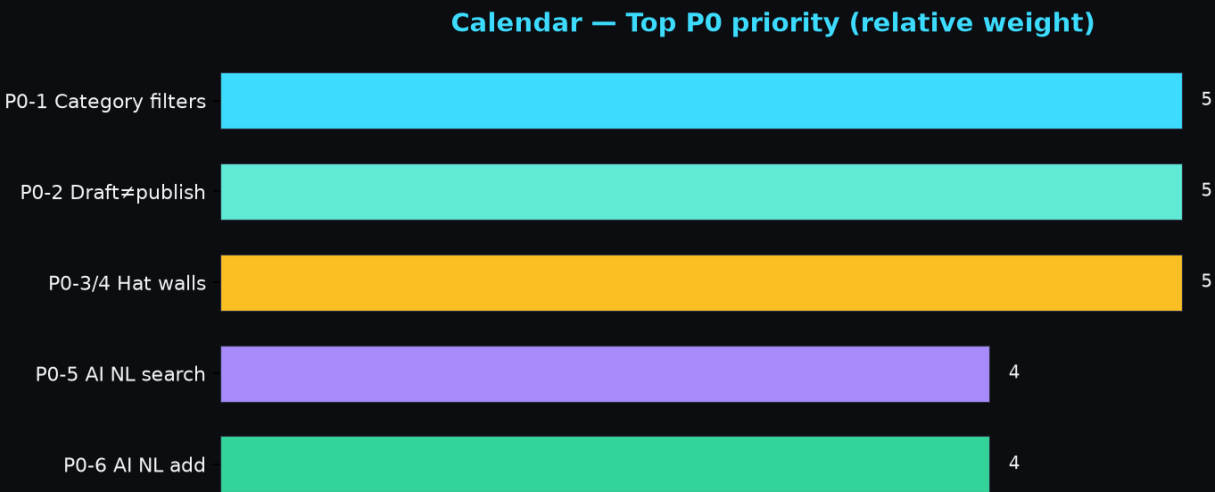


Figure: Top P0 priority weights from the digest.

Calendar — Research + Recommended Features Digest (R2)

Date: 2026-09-24 Author: Chief of Staff / Grok Bot (Kelyra) Status: Ready for Chuck review SoT pair: [notes/company/calendar-research.md](#) · [notes/company/calendar-plan.md](#) Also keep (unchanged architecture/security/acceptance unless noted): sibling [*-architecture.md](#) / [*-security.md](#) / [*-acceptance.md](#)

This dated file is a readable pack for Chuck. Full detail lives in the SoT pair above.

Top 5 P0 recommendations

1. P0-1 — Category filters for full CEO taxonomy (school, class, lesson topic, assignment, quiz, study, test, project, sport, personal, absence) on the existing Calendar surface (not a greenfield app).
2. P0-2 — Draft due dates never auto-publish; teacher Needs/To-Do Publish to calendar (pop quiz law) completed end-to-end.
3. P0-3/4 — Hat visibility walls + twin/focused-child fail-closed (filters ≠ security).
4. P0-5 — AI NL search/filter over the already-visible set only.
5. P0-6 — AI NL add with preview + Save (parent doctor appointment → that child's teachers only; never school firehose; never twin mix).

Research (full SoT)

Calendar Research Note (CAL-R2)

Date: 2026-09-24 Author: Chief of Staff / Grok Bot (Kelyra) Status: Ready for Chuck review Revision: R2 (supersedes CAL-R1 / 2026-09-03 thin refresh of deep-research-9) Cards: Epic [t_908912d4](#) · related architecture/security/acceptance remain SoT for impl gates Access date for all URLs below: 2026-09-24 (America/Chicago)

Related (do not rebuild): Diary epic (separate) · Lesson plans [notes/company/lesson-plan-research.md](#) · Class landing [notes/company/class-landing-research.md](#) Shipped ground (not greenfield): [calendar-r3-kelyra-current.md](#), [calendar-r5-intent.md](#), [calendar-r5-pm-lock.md](#), [calendar-3d-wheel-intent.md](#), [calendar-architecture.md](#), [calendar-security.md](#), PersonTabs / CR-CalTabs notes.

Executive summary

CEO wants a full school calendar of anything with a date or span: school events, sports, class activities, lesson topics, assignments, quizzes, study work, tests, projects — with category filters, hat walls (5th ≠ 3rd; parents only their kids; twins never mixed), day/week/month (+ research-backed extras), draft due dates that do not auto-publish (pop quiz) + teacher To-Do to publish, and AI NL search/filter + AI add (parent doctor appointment → correct teachers only).

R2 verdict: Kelyra Calendar is not greenfield. R2-R4 + CR-CalTabs already shipped a real Calendar surface (PersonTabs Y/M/W/D, category chips, Calendars sheet, Hidden/assign≠publish laws, hats/twins, AI draft-then-Save composer pattern). R5 polish and 3DW wheel are design-stamped, Eng-gated. The gap vs CEO ask is mostly product completeness (full dated taxonomy, NL search/add, publish To-Do end-to-end, sports opt-in depth) — not inventing a new calendar app.

Copy from competitors: layered calendars + course filters (Canvas, Presence); draft/scheduled hide (Google Classroom); sports opt-in (sportsYou); Day/Week/Month/Agenda (+ Year for orientation, Apple-like). Skip: full Google/Outlook clone; RSVP mesh; anonymous public school calendar; Diary-on-Calendar; office homework

firehose.

1. What already shipped vs still open

Honest inventory grounded in company notes (code-backed R3 inventory + R4/R5/3DW locks). Flag: live DB column names may differ from architecture sketches — verify before Eng; this research does not invent migrations.

Area	Status	Evidence / notes
Calendar route + tray/drawer entry	Shipped	R2–R4 live surface
PersonTabs Y / M / W / D	Shipped	CR-CalTabs one-row
Phone agenda/day defaults; web week emphasis	Shipped	<code>calendar-r3-kelyra-current.md</code>
Category Show chips (Academic / School / Sport / Personal spirit)	Shipped	LF-A HOLD through R5
Calendars nested sheet under Settings	Shipped	stack law refined in R5
Hats / dual-hat seat scope	Shipped (law)	fail-closed
Twins / focused child (CH-A)	Shipped (law)	never unlabeled merge
Hidden dues; assign ≠ publish	Shipped (law)	pop-quiz; Needs publish
AI draft banner → teacher Save	Partial	composer pattern; not full NL search/add
R5 12 chrome locks	Partial / Eng-gated	dual stamp MET; Eng dark until Chuck send
3DW period wheel	Partial / Eng-gated	design stamp; Eng dark
Full <code>calendar_events</code> taxonomy (school/class/sport/personal/absence...)	Partial → Proposed	architecture sketch exists; do not claim full live schema without verify
NL search over visible set	Proposed (P0)	CEO 2026-09-03
NL add (parent doctor appt) with visibility	Proposed (P0)	CEO example
Teacher To-Do “Publish to calendar” completeness	Partial → P0	law exists; UX may still be incomplete
Sports team opt-in depth (roster, coach)	Partial	opt-in law; not sportsYou clone
iCal subscribe / two-way sync	Proposed (P2)	later
Diary on Calendar	Out of scope	separate epic

2. Competitor & pattern analysis (citations)

2.1 Canvas LMS Calendar

- Global calendar aggregates all courses’ assignments + events; sidebar course filter (color boxes); Week / Month / Agenda; personal calendar; iCal feed; Scheduler optional.
- Instructor: create events/assignments; drag-drop dates sync to Assignments/Syllabus/Grades.
- Copy: one surface + per-class filters; personal layer; agenda on phone.
- Skip: Scheduler appointment mesh; 10-calendar hard cap as product law (Canvas default — not ours).
- Sources: What is the Calendar?; Filter by course; How do I use the Calendar?. Accessed 2026-09-24 (America/Chicago).

2.2 Google Classroom + Google Calendar

- Classroom calendar (computer): due dates only; cannot add; guardians with email summaries cannot see Classroom calendar.
- Google Calendar: class calendars; personal reminders stay private; guardians see nothing unless calendar shared.
- Draft / scheduled classwork not visible until posted.
- Copy: draft/scheduled hide until publish; guardians fail-closed by default.
- Skip: relying on external Google Calendar as SoT; guardian-blind calendar as the only parent story (Kelyra parents need a first-class, hat-scoped agenda).
- Source: View due dates and events in a calendar. Accessed 2026-09-24 (America/Chicago).

2.3 PowerSchool Presence calendars

- Day List / Weekly / Monthly / Event List / Yearly views; category filters; search name/description/location/color; merge calendars; optional admin approval before publish; iCal/CSV import-export.
- Copy: multi-view + category filters; optional approval for school-wide publish; merge school + class layers.
- Skip: public-ish CMS calendar as default for student education records; event registration/waitlist mesh in v1.
- Source: Calendars Guide (Presence). Accessed 2026-09-24 (America/Chicago).

2.4 Schoology (PowerSchool L&E)

CEO full dated taxonomy — category filters on existing Calendar

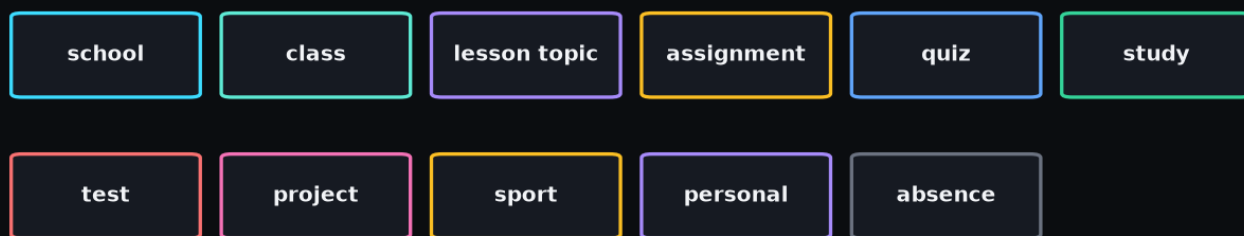


Figure: CEO full dated taxonomy as filter chips.

- Course calendar: month/week/day; assignment/assessment dues appear automatically; homepage tools separate.
- Copy: dues auto-project when published; course home ≠ calendar (join).
- Source: Schoology Calendar. Accessed 2026-09-24 (America/Chicago).

2.5 sportsYou (athletics)

- Team calendars, real-time schedule updates, opt-in teams, school-safe messaging; AD can filter teams by season; Pro adds advanced views / printable filtered calendars.
- Copy: sports = opt-in roster, never school firehose; coach/AD scoped.
- Skip: rebuilding sportsYou messaging inside Calendar.
- Sources: sportsYou; Calendar help collection. Accessed 2026-09-24 (America/Chicago).

2.6 Apple Calendar view patterns (school-relevant UX)

- Day / Week / Month / Year / List; iOS 18 month density modes; pinch density. School apps often mirror Year→Month→Day hierarchy for orientation.
- Copy: Year for orientation (Kelyra R4 HOLD); List/Agenda for phone; Month grid on web.
- Skip: full EventKit two-way sync as v1 SoT.
- Sources: Change how you view events (Apple); iOS 18 Calendar coverage (secondary). Accessed 2026-09-24 (America/Chicago).

2.7 Gap no competitor owns

No dominant K-12 product combines academic + sports opt-in + personal/absence with Hidden pop-quiz, per-child twin walls, and AI NL under the same RLS seat while refusing Diary conflation. That is Kelyra’s cut.

3. Information architecture

Layered calendars on one surface (school / class / academic work / sport / personal / absence), not separate apps.

Layer	Who creates	Default filter	Visibility wall
School	Office	ON all hats	School-wide published
Class activity	Teacher of class	ON for enrolled / taught	Class roster only
Assignment / quiz / test / project / lesson topic	Teacher (via assignment or plan join)	ON when published to calendar	Enrollment + publish gate
Sport	Coach / AD	OFF until opt-in	Team roster
Personal / study	Owner	Owner ON	Self (share later)
Absence / pull-out	Parent (or office attendance later)	Relevant teachers ON	<code>student_teachers</code> of that child only — not school firehose; ...

Filters ≠ security. Chips only subset what RLS already allows (`calendar-security.md`).

4. Hat / visibility matrix (FERPA fail-closed)

Item	Teacher (taught)	Student	Parent (child C)	Office
School event	Y	Y	Y	Y create
Class T event	If teaches T	If enrolled T	If C in T	Directory; not teacher personal
Published due (T)	Teaches T	Assigned/enrolled	C has item	No homework firehose v1
Hidden due (pop quiz)	<code>class_teacher_of</code> only	N	N	N
Sport team	Coach / opted	Roster	C opted	School-marked announcements only
Parent doctor pull-out for C	Teachers of C’s classes	N	Parents of C	N default
Twin sibling	—	—	Never on C’s stream	—

Honest FERPA: Published assignment dues are education-record-adjacent; Hidden stays teacher work product until publish. Kelyra does not claim “school official” status for AI vendors without DPA (`calendar-security.md`). Access date note: ED FERPA overview at studentprivacy.ed.gov — treat as primary; exact FAQ wording not re-fetched this pass if CDN blocked (flag unverified verbatim).

5. Views: research-backed + mobile vs web

View	Research backing	Phone	Web
Agenda / Day List	Canvas Agenda; Apple List; Presence Day List	Primary	Sidebar / mode
Day (timeline)	Universal	Detail / Single mode	Detail
Week (3/5/7)	Teacher planning; R5 locks labels	Secondary	Primary for teachers
Month	Canvas default; Presence Monthly	Compact / List	Full grid
Year	Apple orientation; Kelyra R4 HOLD	Orientation	Orientation
Multiday	Presence / Apple	Later / R5 list covers continuous	Optional

R5 HOLD: Week 3=TUE-THU, 5=MON-FRI, 7=full; range MM/DD/YYYY-MM/DD/YYYY; Day List continuous without date chevron.

6. Draft vs published + teacher To-Do

- Setting a due date must not auto-publish to student/parent calendars (pop quiz).
- Google Classroom pattern: drafts/scheduled hidden until post.
- Presence: optional admin approval for publish to shared calendars.
- Kelyra law: **assign** ≠ **publish**; Hidden until teacher Publish to calendar from Needs / To-Do; re-hide is v1.1 per architecture.
- Unpublished items remain on teacher calendar/To-Do only.

7. AI: NL search + NL add (CEO)

Capability	Behavior	Safety boundary
NL search/filter	Parse over already-visible rows only	Never elevates RLS; twins confirm
NL add	Draft event → preview “who can see this” → Save	Never auto-publish; never create class/student; never Approve grades
Parent example	“Johnny doctor Tuesday 12pm” → absence/pull-out, <code>student_id=Johnny...</code>	Not school firehose; not sibling

Stack ground: Expo client → Supabase RLS → Edge AI adapters. Ops/GTM notes use Gemini on Edge; `docs/architecture.md` still documents xAI adapter — do not invent a stack rewrite in this epic.

Pop-quiz law: draft due dates never auto-publish

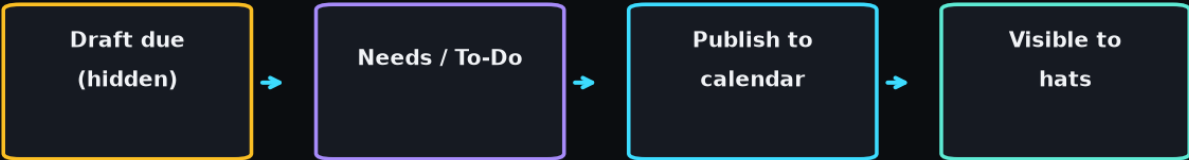


Figure: Draft due → Needs To-Do → Publish flow.

8. Cross-links

Topic	Join rule
Lesson plans	Published plan may occupy a calendar span by reference — not duplicate body (...)
Class landing	Landing shows calendar slice join — not a second calendar
Diary	Never on Calendar; reflection stays Diary epic

9. Unverified / open research flags

- Exact live migration state of `calendar_events` / `calendar_visibility` columns — verify in DB before Eng; architecture is sketch.
- Market share stats for Canvas vs Classroom calendars — not invented; qualitative only.
- Parent satisfaction numbers for twin separation UX — no public primary study found; treat as product law not survey claim.
- sportsYou internal roster API details — not needed; pattern only.
- FERPA FAQ exact wording if ED fetch blocked this pass — use `studentprivacy.ed.gov` as SoT when Eng writes copy.

10. Open questions for Chuck

1. Confirm PO = NL search + NL add + publish To-Do completeness on top of shipped chrome (vs pause for R5/3DW Eng first)?
2. Office: school events only, or any aggregate “counts” without row dump?
3. Sports: thin opt-in calendars now, or wait for AD hat?
4. Parent pull-outs: notify all of C’s teachers always, or teacher-selected classes?
5. iCal subscribe timing (P2) — any Spring Baptist demand now?

RECOMMENDED NEXT ACTION: Chuck reviews R2 research + `calendar-plan.md`; keep architecture/security/acceptance; Eng only after explicit send (not this card).

Plan / recommended features (full SoT)

Calendar Plan (CAL-P2 / R2)

Date: 2026-09-24 Author: Chief of Staff / Grok Bot (Kelyra) Status: Ready for Chuck review Revision: R2 (upgrades CAL-P1 2026-09-03; grounded in shipped Calendar R2-R5/3DW/PersonTabs) Research: `notes/company/calendar-research.md` (CAL-R2) Keep as SoT (still fit): `calendar-architecture.md`, `calendar-security.md`, `calendar-acceptance.md` — this plan points to them; does not replace RLS law. Stack ground: Expo (iOS/Android/web) + Supabase (Auth/Postgres/RLS/Storage/Edge) + Edge AI adapters. Ops notes currently emphasize Gemini on Edge; `docs/architecture.md` still documents xAI adapter — Calendar must not invent a rewrite.

Audience: Chuck / CoS. Not an Eng send. No app/TS/SQL from this card.

0. Product law (one screen)

Surface	Job	Not
Calendar	One dated/span surface of school · class · academic work · sport · personal · absence the active seat m...	Not Google Calendar clone; not Diary; not office homework firehose
Assignment dues	Grade-book due_at projected when calendar-published	Not auto-broadcast of every planned column
Events	First-class dated/span rows with explicit visibility	Not Feed; not Messages; not audit log
Ask / AI	NL search visible set; NL draft add → confirm → Save	Not diary; not class-create; not grade Approve; not twin merge

CEO bar: full dated school calendar + category filters + hat walls + draft≠publish + AI NL search/add (doctor example).

1. Shipped / Partial / Proposed (feature inventory)

1.1 Shipped (build on; do not re-spec as greenfield)

ID	Feature	Notes
S1	Calendar surface (mobile + web)	Live post R2–R4
S2	PersonTabs Y / M / W / D	CR-CalTabs
S3	Category Show chips	Academic/School/Sport/Personal spirit; filters ≠ security
S4	Settings → Calendars nested sheet	R5 stack law if Eng-sent
S5	Hats + dual-hat seat scope	Fail-closed
S6	Twin / focused-child wall (CH-A)	Never unlabeled merge
S7	Hidden dues; assign ≠ publish	Pop-quiz law
S8	Lean event composer + AI draft-then-Save pattern	Partial AI; Save required
S9	Sports opt-in read concept	Not sportsYou
S10	Day Single vs List modes	R4/R5

1.2 Partial / Eng-gated

ID	Feature	Notes
X1	R5 12 chrome locks	Dual stamp; Eng dark until Chuck send
X2	3DW period wheel	Design stamp; Eng dark
X3	Hybrid projection (due_at + c...	Architecture SoT; verify DB
X4	Needs “Publish to calendar” To-Do UX completeness	Law exists

1.3 Proposed (R2 priority)

Pri	ID	Feature	Rationale
P0	P0-1	Category filters covering CEO taxonomy (school, class, lesson topic, assignment, quiz, study, test, project, s...	CEO must-have
P0	P0-2	Draft due dates never auto-publish + teacher To-Do publish/unhide	Pop quiz
P0	P0-3	Hat visibility matrix enforced in read model (student/parent/teacher/office)	FERPA fail-closed
P0	P0-4	Twins never mixed; parent requires focused child	Product law
P0	P0-5	AI NL search/filter over visible items	CEO 2026-09-03
P0	P0-6	AI NL add with preview + Save (parent doctor → student_teachers only)	CEO example
P1	P1-1	Finish R5 chrome + 3DW when Chuck sends	Polish on live
P1	P1-2	Lesson-plan span join (reference id + span)	Cross-epic
P1	P1-3	Class-landing calendar slice join	Cross-epic
P1	P1-4	Sport team roster depth (coach create, opt-in)	Thin, not sportsYou
P1	P1-5	Recurring school events (simple)	Presence-like
P2	P2-1	iCal subscribe (read-only)	Canvas pattern; later
P2	P2-2	Two-way Google/Apple sync	Explicit non-goal for v1
P2	P2-3	RSVP / attendance on events	sportsYou depth later
P2	P2-4	Public anonymous school calendar URL	FERPA; signed-in only v1

2. Needed vs Desired

Needed (P0)	Desired (P1-P2)
Full dated taxonomy on one surface	Rich sports RSVP
Filters + hat walls + twin wall	iCal / external sync
Draft#publish + Publish To-Do	Office homework aggregates
Day/Week/Month (+ Year/Agenda already in spine)	Advanced recurrence
AI NL search + AI draft add	Auto-suggest from syllabus

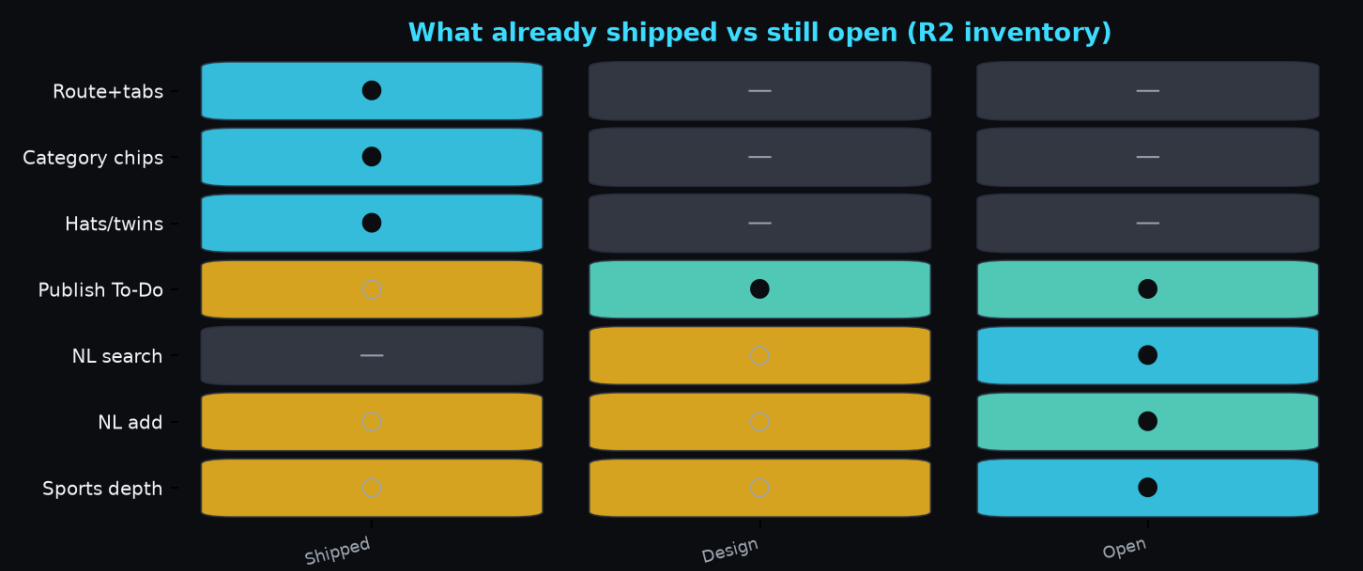


Figure: Shipped vs design vs open inventory.

3. Hat matrix (product)

Hat	Can create	Can see	Cannot
Teacher	Class events; assignments; own study; publish/unhide taught-class dues	Taught classes + school + opted sport + incoming absences for roster	Other teachers' personal; Hidden of non-taught; twin merge N/A
Student	Own study/personal thin	Enrolled published + school + opted sport	Hidden quizzes; other students; sibling
Parent	Absence/pull-out for linked child; own personal	Focused child's published + school + child's sport	Other child unlabeled; Hidden; school homework firehose
Office	School events	School + directory class events (not teacher personal; not hidden quizzes; not parent doctor notes)	Using calendar to mint students/classes

Teachers do not create classes. Calendar does not widen `is_staff`.

4. UI/UX – mobile vs web

	Mobile (Expo)	Web (Expo web)
Default	Agenda / Day List	Week (teachers) / Month (office)
Nav	PersonTabs Y/M/W/D; tray Calendar	Same tabs; denser grids
Filters	Chip sheet / gear	Persistent chip row + Calendars
Create	Sheet composer	Modal composer
AI	Ask sheet → draft card → Save	Same; keyboard-first
Density	List-first; avoid dense month as home	Month/week grids OK

HOLD: Diary ≠ Calendar; Desk ≠ Year; filters ≠ security; R5 title first-tap; Clear Filters = deselect all.

5. AI safety boundaries

1. Search visible set for active seat.
2. Add = draft → “who sees this” preview → explicit Save.
3. Never auto-publish. Never create class/student/link. Never Approve grades.
4. Twin ambiguity → confirm picker.
5. No health detail in logs (ids not “doctor for ADHD...” bodies).
6. No “Ask as superuser.”

6. Implementation approach (no code here)

Grounded in `calendar-architecture.md` / `calendar-security.md`:

1. Hybrid SoT: keep `assignments.due_at` as grade due; project when calendar-published; do not duplicate dues into events.
2. `calendar_events` for school/class/sport/personal/absence spans.
3. `list_calendar_items` SECURITY DEFINER with seat predicates inside; client never wide-selects then filters.
4. Hidden: `class_teacher_of` only — do not reuse office `teaches_class` bypass for family reads.
5. Ask tools: `calendar_search` / `calendar_draft_event` as new calendar.* permissions — do not hang on `assignments.manage`.
6. Ship increments: (A) publish To-Do completeness + taxonomy chips → (B) NL search → (C) NL add → (D) sport depth → (E) iCal.
7. Join lesson-plan spans + landing slice by reference, not copy.
8. Stack: Expo UI + Supabase RLS + Edge AI; reuse existing Calendar chrome.

Gate: No Eng until Chuck says send on a build card. This epic card is research/plan only.

7. Cross-links

- Research: `calendar-research.md`
- Architecture / security / acceptance: keep; refresh cross-links only if stale stubs
- Lesson plans: `lesson-plan-plan.md` § calendar span
- Class landing: `class-landing-plan.md` § calendar slice
- Diary: out of scope

8. Open questions for Chuck

1. Send Eng on R5/3DW before or after P0 NL?
2. Parent pull-out: all of child's teachers vs pick classes?
3. Office see any class homework rows ever in v1? (Plan recommends no.)
4. Spring Baptist sports: which teams first?
5. Confirm AI vendor posture copy for calendar NL (Gemini/xAI) stays “no school-official claim without DPA.”

RECOMMENDED NEXT ACTION: Chuck review; then optional PM story split — still no implementation from this card.

Meta · Card `t_908912d4` (CAL) · Dark visual edition 2026-09-24 · Not legal advice where privacy/compliance topics appear · No Hermes staffing.